Evalanime – Your AI Teaching Assistant

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Team Name:

Psychotic-Coders

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Overview

An open-source agentic AI system that grades student assignments end-to-end. A planner agent dynamically selects tools per submission (parsers, OCR, scoring, plagiarism detection, vision analysis) instead of following a fixed pipeline. Built on Agno with Google Gemini 2.5 on Vertex AI; FastAPI backend, React frontend with live execution streaming over WebSocket.

Objectives

- Build a planner-driven agent whose execution path is decided at runtime, not hard-coded.
- Demonstrate capability-aware behavior, the agent reasons about which tool to use and routes to human review when none fits.
- Enforce submission-format compliance as a graded dimension via a deterministic tool.
- Detect plagiarism in two layers: a non-LLM pairwise similarity (TF-IDF + n-gram Jaccard) plus an LLM judgment that uses the deterministic flags as evidence.
- Generate personalized feedback per student and dispatch via SMTP email.
- Provide a live, transparent UI showing every tool call, prompt, and decision, fully auditable.

Architecture

Frontend (React + Vite + Tailwind) talks to a FastAPI backend over REST + WebSocket. The backend runs an Agno multi-agent team in-process. Each agent has a focused tool registry; the Coordinator picks which specialist to invoke next based on run state. Storage is filesystem-backed (data/assignments, data/runs) with a SQLite audit log. Authentication uses Application Default Credentials required by the institutional Workspace policy that disallows API keys.

Agent Team

Agent	Role
Coordinator (Flash)	Plans the run, routes work to specialists, decides skips/retries.
Capability Inspector (Pro)	Reads the assignment; classifies questions as text vs visual; identifies risks.
Rubric Designer (Pro)	Drafts a detailed rubric with weighted criteria and level descriptors.
Submission Grader (Flash)	Picks parser per submission; computes format compliance; scores against rubric.
Plagiarism Investigator	Runs deterministic similarity + LLM corroboration; flags suspicious pairs.
Reporter (Flash)	Composes per-student feedback; dispatches via Gmail SMTP.

Key Features

- Dynamic tool selection per submission (PDF / DOCX / ZIP / scanned-image PDFs).
- Tesseract OCR fallback when extracted text is sparse.
- Gemini 2.5 multimodal vision for diagram / visual questions.
- Format-compliance deductions enforced deterministically (not by the LLM).
- Pairwise similarity heatmap (TF-IDF cosine + 5-gram Jaccard) catches student-vs-student copying.
- LLM plagiarism judgment uses similarity flags as evidence never the sole basis for an accusation.
- Personalized feedback emails dispatched via Gmail SMTP (app password).
- Live phase tracker, expandable agent trace (full prompts + responses), per-student results, and rubric viewer in the UI.
- Complete audit trail (events.jsonl + summary.json) per run for inspection / contestability.

Technology Stack

Agno 2.6 · Gemini 2.5 Pro & Flash on Vertex AI (ADC) · FastAPI · React 18 + Vite + Tailwind · TanStack Query · pypdf, python-docx, Tesseract · scikit-learn (TF-IDF) · smtplib (Gmail SMTP) · SQLite + filesystem storage.

Timeline

Day	Focus	Deliverable
1	Foundations	Backend + frontend scaffold; ADC auth; demo data generator with 6 varied synthetic submissions.
2	Agent layer	Agno team with all tools; WebSocket streaming; live phase tracker UI.
3	Polish	SMTP email; vision; results & rubric panels; click-through phase drilldown; demo rehearsal.

Evaluation

Evaluated on six synthetic students whose submissions are designed to trigger specific agent capabilities:

Student	Pattern	Capability tested
1	Ideal PDF	Gold-standard path
2	DOCX	Format-compliance deduction (-5)
3	PDF, paraphrased copy of #2	Non-LLM similarity flag
4	ZIP archive	Unzip + nested-PDF discovery
5	Scanned-image PDF	Sparse-text fallback to OCR
6	Partial PDF, missing diagram	Visual-question handling

Future Work

Hosted multi-tenant SaaS pending Google OAuth verification and a third-party CASA security assessment for restricted Classroom + Gmail scopes. Vision-based grading of hand-drawn diagrams. Code-execution sandbox for programming assignments. Cross-semester plagiarism corpus.